



ViWiQA: Efficient end-to-end Vietnamese Wikipedia-based Open-domain Question-Answering systems for single-hop and multi-hop questions

Dieu-Hien Nguyen ^{*,1}, Nguyen-Khang Le ¹, Le-Minh Nguyen ²

Japan Advanced Institute of Science and Technology, Nomi, Japan

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ABSTRACT

Open-domain Question-Answering (QA) task requires a QA system to answer a given question using a large knowledge base like Wikipedia. Modern Open-domain QA systems often follow the two-stage framework Retriever-Reader where the retriever greatly impacts the end-to-end performance. Efficient Vietnamese Open-domain QA systems for single and multi-hop questions have yet to be studied. Although resource-rich languages like English witnessed many advancements in Open-domain QA, these methods often suffer from low data situations. This study proposes ViWiQA, an efficient Vietnamese Open-domain QA system over the Wikipedia knowledge base, with two novel retriever methods for single-hop and multi-hop questions. ViWiQA can be effectively trained with low data and significantly outperforms Lucene-BM25 and Dense Passage Retrieval when adapted to Vietnamese datasets. For single-hop QA, the proposed retriever outperforms Lucene-BM25 by 20% in top-1 retrieval accuracy, and the end-to-end system achieves 15% and 17% absolute gain in EM and F1 scores, respectively. For multi-hop QA, the proposed retriever increases the accuracy of retrieving correct passage pairs by 4% compared to Lucene-BM25, and the end-to-end system shows 7% and 17% absolute gains in EM and F1 scores.

1. Introduction

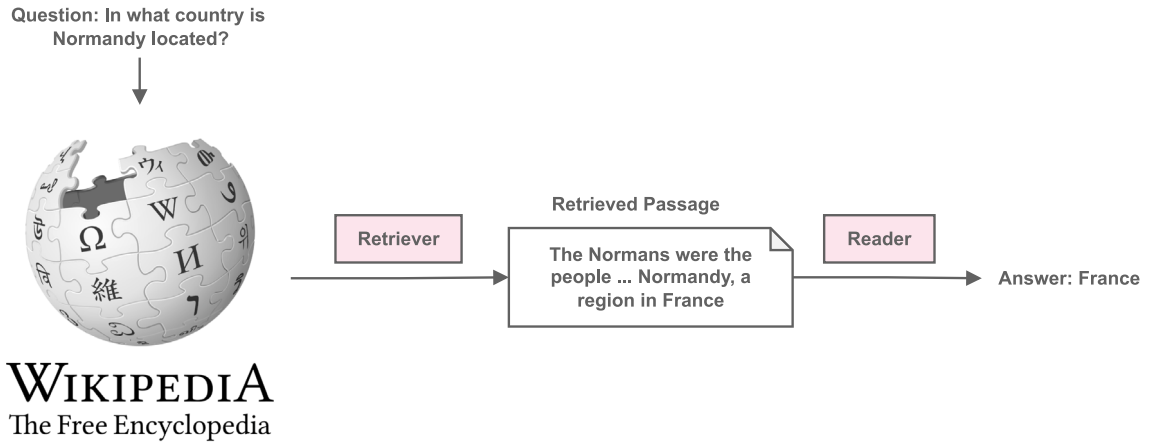
Open-domain QA is an essential QA task requiring the system to input a question and seek the answer using a knowledge base. Early QA approaches are often sophisticated due to many components constituting the system. As deep learning progresses, more recent methods take advantage of Machine Reading Comprehension (MRC) approaches and simplify the system into a framework consisting of two components: Retriever and Reader. For a query, the Retriever aims to retrieve relevant documents from the knowledge base, and the Reader aims to find the answer using those documents. Advancements in Open-domain QA are witnessed in resource-rich languages like English, with many retriever methods proposed. While sparse retrievers like TF-IDF or BM25 were used in early QA systems (Chen, Fisch, Weston, & Bordes, 2017), retriever approaches using dense representations (Gao & Callan, 2021; Guu, Lee, Tung, Pasupat, & Chang, 2020; Karpukhin et al., 2020; Lee, Chang, & Toutanova, 2019; Ren et al., 2021) produced competitive results and became a new paradigm for passage retrieval in Open-domain QA. Using a reranker to rerank the retrieval result was also a popular technique in Information Retrieval (IR) (Nogueira & Cho, 2019; Nogueira, Jiang, Pradeep, & Lin, 2020;

* Corresponding author.

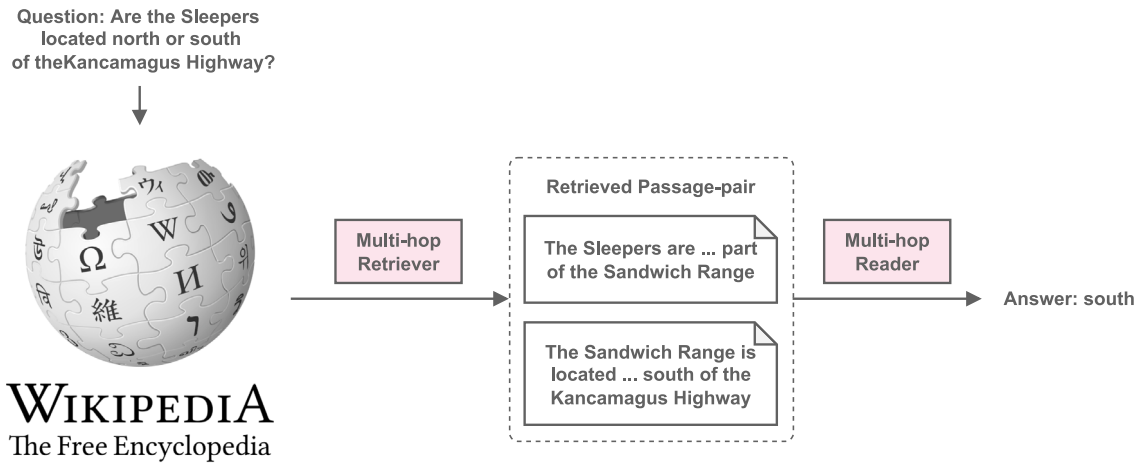
E-mail addresses: ndhien@jaist.ac.jp (D.-H. Nguyen), lnkhang@jaist.ac.jp (N.-K. Le), nguyenml@jaist.ac.jp (L.-M. Nguyen).

¹ These authors contributed equally to this work.

² Co-ordinator and supervisor.



(a) Pipeline of Open-domain single-hop QA system with an example question from SQuAD dataset



(b) Pipeline of Open-domain multi-hop QA system with an example question from HotpotQA dataset

Fig. 1. General pipeline of Open-domain single-hop and multi-hop QA systems.

Pradeep, Nogueira, & Lin, 2021; Qiao, Xiong, Liu, & Liu, 2019) and Open-domain QA (Lee, Yun, Kim, Ko, & Kang, 2018; Mao et al., 2020; Wang et al., 2018). Many datasets for QA and Open-domain QA were proposed in English (Dunn et al., 2017; Joshi, Choi, Weld, & Zettlemoyer, 2017; Kwiatkowski et al., 2019; Rajpurkar, Zhang, Lopyrev, & Liang, 2016). Fig. 1(a) illustrates a system of Open-domain single-hop QA with a general pipeline. Multi-hop QA, which requires relating pieces of information from multiple documents, was also proposed. Yang et al. (2018) proposed HotpotQA as a large-scale multi-hop dataset and provided benchmarks for QA with given contexts and Open-domain QA. Multi-hop retrievers that aim to retrieve the passage pairs to perform multi-hop reasoning were proposed (Asai, Hashimoto, Hajishirzi, Socher, & Xiong, 2020; Xiong et al., 2021; Zhu, Pang, Lan, Shen, & Cheng, 2021). Moreover, multi-hop readers have also been actively studied (Alkhaldi, Chu, & Kurohashi, 2021; Li, Lei, & Yang, 2022; Shao, Cui, Liu, Wang, & Hu, 2020; Yuntao, Phuong, Racharak, Le, & Minh, 2022). Fig. 1(b) shows a general pipeline of Open-domain multi-hop QA systems.

Despite these advancements, Vietnamese, a language with over 98 million native speakers worldwide, remains a low-resource language with a scarcity of research on QA systems. Only a handful of datasets and studies were proposed for Vietnamese QA. UIT-ViQuAD (Nguyen, Nguyen, Nguyen, & Nguyen, 2020) and VIMQA (Le, Nguyen, Le Thanh, & Nguyen, 2022) are two large-scale Vietnamese datasets for single and multi-hop QA, respectively. Nguyen et al. (2022) proposed a Vietnamese Open-domain single-hop QA system XLMRQA. However, XLMRQA only uses a simple sparse retriever, and its knowledge base is about 800 times smaller than the Vietnamese Wikipedia regarding the number of passages. To our knowledge, no efficient Vietnamese Open-domain QA system for single and multi-hop questions has been proposed. Also, no QA system that uses the entire Vietnamese Wikipedia as the knowledge base has been proposed.

Although many state-of-the-art methods in Open-domain QA achieved outstanding performances in English QA benchmarks, their capabilities are not guaranteed when adapted to a low-resource language like Vietnamese. Retriever approaches using dense

representation like Dense Passage Retrieval (DPR) (Karpukhin et al., 2020; Xiong et al., 2021) were shown to require much data and complex techniques for efficient training and perform poorly in low data circumstances (Gao & Callan, 2021; Ren et al., 2021). This behavior can be observed in our experiments when adapting DPR to Vietnamese datasets. While Cross-Encoder rerankers (Devlin, Chang, Lee, & Toutanova, 2019; Yang, Dai, et al., 2019) are more effective than dense retrievers in low data situations, they are impractical when used with a large number of documents in a corpus regarding one question. Moreover, it is not trivial to derive the training data for Cross-Encoder from a QA dataset. Based on related works, we believe that no effective retriever method has been suggested for single or multi-hop Vietnamese Open-domain QA. The reader component also plays an essential role in Vietnamese QA systems. Common Vietnamese reader approaches (Nguyen et al., 2022; Nguyen, Nguyen, Do, Nguyen, & Nguyen, 2021) used the pre-trained multilingual model XLM-RoBERTa (Conneau et al., 2020) and showed potential results. No study has been conducted to compare reader approaches using different pre-trained models on Vietnamese corpus like PhoBERT (Nguyen & Tuan Nguyen, 2020), and multilingual corpus such as Multilingual-BERT, XLM-RoBERTa.

Entity resolution (ER) task requires recognizing entities corresponding to the same entity in the real world across different data sources. Regarding End-to-end Open-domain QA systems, the entity that can answer the question might have different aliases across the knowledge base. For example, on Wikipedia, “Edson Arantes do Nascimento” and “Pelé” can refer to the same professional footballer. This creates inconsistency in the responses of QA systems, where the answers of the systems vary depending on the retrieved contexts and knowledge bases. Moreover, answers from the QA system can be ambiguous and refer to multiple entities. For example, in Wikipedia 2022, the entity “James Abbott” referred to 8 different persons, and the entity “La Villa” referred to 8 different places. It is essential that the QA systems can provide the user with the correct entity. However, the problem of ER in Open-domain QA is overlooked, and no research has been conducted on the Vietnamese knowledge bases. The contest Zalo AI Challenge 2022³ introduced the End-to-end QA task over Wikipedia, in which the input is a plain text question, and the answer can be a Wikipedia entity, a specific date, or a number. This task posed many challenges to existing QA approaches.

To address the mentioned challenges in Vietnamese Open-domain QA, this study proposes the ViWiQA system. The core idea is to overcome the lack of data in the low-resource language Vietnamese by employing and combining architectures that can be effectively trained with a limited amount of data and leveraging the structural meta-data of Wikipedia, such as the hyperlink graph and redirect pages. Specifically, the single-hop retrieval module integrates the Cross-encoder and span-extraction Reader architectures, built on top of the Transformer model with additional regression and span-prediction layers. The retrieval process involves filtering the results obtained from BM25 using the Reader and reranking by the Cross-encoder to obtain the final results. The multi-hop retrieval module also builds upon the single-hop module for passage retrieval and utilizes the Wikipedia graph to identify candidates for subsequent hop retrieval. Moreover, the system also addresses the ER problem in QA by employing the redirect pages meta-data from Wikipedia. As a result, ViWiQA can be effectively trained in a low-resource language like Vietnamese and achieve high performance in single-hop and multi-hop Vietnamese QA benchmarks, obtaining the Runner-up prize in Zalo AI Challenge 2022.

2. Research objectives

This study proposed new state-of-the-art multilingual retriever methods for single and multi-hop Open-domain QA that can be efficiently trained with low resources. Using these retrievers, we proposed **ViWiQA**, a set of efficient end-to-end Vietnamese Open-domain QA systems taking Wikipedia as their knowledge base. ViWiQA consists of **ViWiQA-Single** and **ViWiQA-Multi** systems, for single and multi-hop QA, respectively. Our code, models, Lucene-BM25 and DPR indexing, and the Vietnamese Wikipedia hyperlink graph are accessible to the public⁴ to support other research about Vietnamese QA. This paper has four main contributions listed below:

1. We propose retriever methods for single-hop and multi-hop Vietnamese Open-domain QA that can be efficiently trained with low resources and establish state-of-the-art retrieval accuracy.
2. Using the proposed retrievers and an effective reader model, we propose the Vietnamese QA system ViWiQA that establishes the new state-of-the-art performance on single and multi-hop Vietnamese QA benchmarks.
3. We propose a simple approach to tackle ER in Wikipedia knowledge bases and develop a complete end-to-end QA system in Vietnamese Wikipedia, which outputs specific Wikipedia entities instead of plain text answers. The approach achieved **Runner-up Prize** in the Zalo AI Challenge 2022 Contest.
4. The ablation study analyzes essential aspects when adapting state-of-the-art English methods to low-resource situations like Vietnamese. The adaptation of these methods and the proposed ViWiQA make a solid foundation for future research in Vietnamese and multilingual Open-domain QA.

3. Related work

The Retriever-Reader pipeline is among the popular techniques for Open-domain QA (Zhu, Lei, et al., 2021). Retrievers that use sparse vectors (TF-IDF, BM25) are called sparse retrievers and are widely used for many QA tasks (Yang, Xie, et al., 2019). Retrievers that represent the questions and the documents as dense vectors via dual-encoders are called dense retrievers (Gao & Callan, 2021; Guu et al., 2020; Karpukhin et al., 2020; Ren et al., 2021). Dense retrievers using two separate encoders for passages

³ An annual AI competition in Vietnam (challenge.zalo.ai).

⁴ Code is available at <https://github.com/ViWiQA/ViWiQA>

Table 1

Example from UIT-ViQuAD dataset. English translation is provided in *italic*. Answer is highlighted in **red**.

Question: Vị trí địa lý của Paris có gì đặc biệt? (*What is special about the geographic location of Paris?*)

Answer: nằm ở điểm gặp nhau của các hành trình thương mại đường bộ và đường sông (*lies at the meeting point of land and river trade routes*)

Context: (from Wikipedia) Paris nằm ở điểm gặp nhau của các hành trình thương mại đường bộ và đường sông, và là trung tâm của một vùng nông nghiệp giàu có. Vào thế kỷ 10, Paris đã là một trong những thành phố chính của Pháp cùng các cung điện hoàng gia, các tu viện và nhà thờ. Từ thế kỷ 12, Paris trở thành một trong những trung tâm của châu Âu về giáo dục và nghệ thuật. Thế kỷ 14, Paris là thành phố quan trọng bậc nhất của Cơ Đốc giáo và trong các thế kỷ 16, 17, đây là nơi diễn ra Cách mạng Pháp cùng nhiều sự kiện lịch sử quan trọng của Pháp và châu Âu. Đến thế kỷ 19 và 20, thành phố trở thành một trong những trung tâm văn hóa của thế giới, thủ đô của nghệ thuật và giải trí.

Translation: Paris lies at the meeting point of land and river trade routes, and is the center of a rich agricultural region. By the 10th century, Paris was already one of France's main cities with royal palaces, monasteries and churches. From the 12th century, Paris became one of Europe's centers for education and the arts. In the 14th century, Paris was the most important city of Christianity and in the 16th and 17th centuries, it was the site of the French Revolution and many important historical events of France and Europe. By the 19th and 20th centuries, the city became one of the cultural centers of the world, the capital of the arts and entertainment.

and questions are commonly used for single-hop (Karpukhin et al., 2020) and multi-hop (Xiong et al., 2021). Recent works have been proposed to improve dense retrievers and make them able to scale effectively to large corpora (Khattab, Potts, & Zaharia, 2021a, 2021b; Khattab & Zaharia, 2020; Santhanam, Khattab, Potts, & Zaharia, 2022; Santhanam, Khattab, Saad-Falcon, Potts, & Zaharia, 2022). Li, Li, and Nie (2022a) proposed DGRCoQA containing retriever and reader components where the retriever is enhanced by graph reasoning. Moreover, retrievers and readers working on multi-modal knowledge sources are also studied. Li, Li, and Nie (2022b) propose MMCQA, a task of multi-modal QA, and a baseline model consisting of a multi-modal retriever and a reader.

Another area of research concentrates on reranking the passages retrieved in the first-stage retriever. Cross-Encoder (Devlin et al., 2019) used for Rerankers in Open-domain QA has shown substantial enhancement (Nogueira & Cho, 2019). Methods using generative models (Nogueira et al., 2020; Pradeep et al., 2021) or integrating retrieval/reranking phases (Karpukhin et al., 2020; Zhang, Long, Xu, & Xie, 2022) also show high performance. Moreover, reranking passages using predictions from the reader model shows potential improvement (Mao et al., 2020).

The reader component can be categorized into Extractive Reader, which aims to extract the most likely spans from given documents as the answers, and Generative Reader, which aims to generate the answers via sequence-to-sequence methods. While earlier QA systems often employ Extractive Reader (Guu et al., 2020; Karpukhin et al., 2020), some recent systems use Generative Reader (Lewis et al., 2020; Xiong et al., 2021).

The Retriever-Reader pipeline is also widely used in multi-hop Open-domain QA. For multi-hop questions, the model needs to perform reasoning over multiple documents. As a result, the retriever component typically retrieves multiple passages instead of one. In the HotpotQA dataset (Yang et al., 2018), the questions require reasoning over two passages. In many multi-hop QA systems (Asai et al., 2020; Xiong et al., 2021; Yang et al., 2018), the number of passages retrieved by the retriever is also two.

Research in cross-lingual and multilingual information retrieval is also actively studied. Asai, Yu, Kasai, and Hajishirzi (2021) proposed a cross-lingual dense retriever for obtaining documents across different languages for a question. Ren et al. (2022) proposed a method combining the sparse and dense models to generate data from the unlabeled corpus for self-supervised training of multilingual retrieval models.

Vietnamese is a low-resource language and suffers from a deficiency of research in QA and Open-domain QA. Only a handful of Vietnamese QA datasets and systems were proposed. UIT-ViQuAD (Nguyen et al., 2020) for single-hop QA and VIMQA (Le et al., 2022) for multi-hop QA are two large-scale Vietnamese datasets. Table 1 shows an example from the ViQuAD dataset, and Table 2 shows an example from the VIMQA dataset. A few Vietnamese QA systems were proposed. ViReader (Nguyen et al., 2021) is an MRC system with given contexts. XLMRQA (Nguyen et al., 2022) is a system for Open-domain QA using all passages in ViQuAD as the knowledge base. ViQAS (Nguyen, Do, Nguyen, Nguyen, & Nguyen, 2023) is a multi-stage transfer learning model based on BERT consisting of several components such as pre-processing rules, text retriever, automatic reader, and answer selection. Although ViQAS shows competitive performance on ViQuAD, the pre-defined set of rules may prevent the model from generalizing beyond the evaluated datasets.

To our knowledge, no effective retriever method for single and multi-hop Vietnamese Open-domain QA has been researched. Also, no Vietnamese QA systems that use large-scale knowledge bases like the entire Wikipedia have been studied, despite their importance in real-life systems.

4. Method

4.1. Problem formulation

The Vietnamese Open-domain QA problem can be explained as follows. Given a question q in the Vietnamese natural language, a QA system must answer the question using a knowledge base C . The knowledge base C contains c passages, denoted as $P_1, P_2, \dots, P_c$,

Table 2

Example from VIMQA dataset. English translation is provided in *italic*. Answer is highlighted in **red**. Supporting facts are highlighted in **blue**.

Question: Biệt danh của một trong những câu lạc bộ ở Anh mà Fabien Alain Barthez từng chơi và đoạt huy chương là gì? (*What is the nickname of one of the clubs in England where Fabien Alain Barthez played and won a medal?*)

Answer: Quỷ đỏ (*The Red Devils*)

Title 1: Manchester United F.C.

Passage 1: (*from Wikipedia*) Câu lạc bộ bóng đá Manchester United (tiếng Anh: Manchester United Football Club, hay ngắn gọn là MU hay Man Utd) là một câu lạc bộ bóng đá chuyên nghiệp có trụ sở tại Old Trafford, Đại Manchester, Anh. Câu lạc bộ đang chơi tại Giải bóng đá Ngoại hạng Anh, giải đấu hàng đầu trong hệ thống bóng đá Anh. Với biệt danh "Quỷ Đỏ", câu lạc bộ được thành lập dưới tên Newton Heath LYR Football Club vào năm 1878, đổi tên thành Manchester United vào năm 1902 và chuyển đến sân vận động hiện tại, Old Trafford, vào năm 1910.

Translation: *Manchester United Football Club (English name: Manchester United Football Club, or simply MU or Man Utd) is a professional football club based at Old Trafford, Greater Manchester, England. The club is playing in the English Premier League, the top league in the English football system. Nicknamed the "The Red Devils", the club was founded as Newton Heath LYR Football Club in 1878, renamed Manchester United in 1902, and moved to its current stadium, Old Trafford, in 1910.*

Title 2: Fabien Barthez

Passage 2: (*from Wikipedia*) Fabien Alain Barthez (sinh ngày 28 tháng 6 năm 1971) là một cựu cầu thủ bóng đá người Pháp. Ông đã từng đoạt một số huy chương khi chơi ở vị trí thủ môn cho Marseille, Manchester United và cùng với đội tuyển bóng đá quốc gia Pháp giành chức vô địch tại World Cup 1998, Euro 2000 và lọt vào trận chung kết World Cup 2006. Ông cùng với Peter Shilton là 2 thủ môn giữ kỷ lục giữ sạch lưới nhất trong giải vô địch bóng đá thế giới, trong 10 trận. Ở câu lạc bộ, ông đã từng chiến thắng tại giải Cúp các đội vô địch bóng đá quốc gia châu Âu, một số danh hiệu tại Giải vô địch bóng đá Pháp và Giải bóng đá Ngoại hạng Anh.

Translation: *Fabien Alain Barthez (; born 28 June 1971) is a French former footballer. He won several medals while playing as a goalkeeper for Marseille, Manchester United and won the 1998 World Cup, Euro 2000, and reached the 2006 World Cup final with the France national football team. He and Peter Shilton are the two goalkeepers who hold the record for keeping the cleanest sheets in the World Cup, in 10 matches. At the club, he won the UEFA Champions League, and several titles at the French Football Championship and the English Premier League.*

Table 3

Description of the notations used in the study.

Scope	Symbol	Description
Single hop	[CLS]	Special classification token whose final hidden state is treated as the representation for classification/regression tasks
	[SEP]	Special token to separate the question and the passage
	Tok i	Token i th of the question/paragraph after tokenization
	E_i	Input embedding corresponding to token i th
	C	Final hidden state corresponding to the [CLS] token
	T_i	Final hidden state corresponding to the i th input token in the question
	T'_i	Final hidden state corresponding to the i th input token in the paragraph
	P_i	Passage i th in the retrieval result
	a_i	Answer in the list of predicted answer by Reader model
	R_i	Passage in the final retrieval result
Multi hop	P_i	Passage in the 1st retrieval result
	Q_i	Passage in the 2nd retrieval result
	A_i	Article in the Wikipedia corpus
	$A_{i,j}$	Article j th that is connected to article A_i in the hyperlink graph
	$P_{i,j,k}$	Passage k th in article $A_{i,j}$
Reader	S	Learnable vector added to calculate the start logits of a span
	E	Learnable vector added to calculate the end logits of a span
	P_{start_i}	The probability of token i th the answer span
	P_{end_i}	The probability of token i th being the end of the answer span

in which passage P_i consists of tokens $p_i^{(1)}, p_i^{(2)}, \dots, p_i^{(l)}$ with l is the length of the passage. For single-hop questions, the task is to retrieve one passage P_i among c candidate passages and use P_i to obtain the answer. For multi-hop questions, the task is to retrieve a pair of passage P_i, P_j among c candidate passages and connect P_i and P_j to get the answer to the question q . Table 3 outlines the symbols used in this study and their meanings.

4.2. Vietnamese wikipedia pre-processing

This section describes building the Vietnamese Wikipedia knowledge base for Open-domain QA and the Wikipedia hyperlink graph. The Vietnamese Wikipedia dump from January 20, 2022, is used as the source document. Inspired by the approach of Chen

et al. (2017), WikiExtractor was employed to get the text-only portion of the Wikipedia dump, removing semi-structured data like lists and tables. Upon obtaining the texts of all articles, a sliding window technique with window size $W = 100$ and stride $S = 50$ is employed to separate the articles into overlapping text chunks, each of which contains W words, following the work of Wang, Ng, Ma, Nallapati, and Xiang (2019). These text blocks are considered passages and used as basic units for retrieval. This process results in 3,885,030 passages.

The Wikipedia hyperlink graph is extracted along with the Vietnamese Wikipedia passages. The Wikipedia hyperlink graph is a directed graph where each node is a Wikipedia article, and each edge (u, v) indicates there exists a hyperlink from the article u to article v . WikiExtractor is employed to get the text portion of the Wikipedia articles, preserving the HTML tags for hyperlinks. A regular expression is then applied to extract the set of linked titles from the HTML tags. For each article u , the set of linked titles V is extracted. For each linked title v_i in V , if there exists an article with the same title v_i after normalization, the edge (u, v_i) is added to the graph.

4.3. Retriever for single-hop QA

We propose a retriever method for Vietnamese single-hop Open-domain QA, namely *ViWiQA-Single Retriever*. Fig. 2 shows the process of the proposed method. Given a question, we first use Lucene-BM25 (Lin et al., 2021) to retrieve top m passages from the Wikipedia knowledge base. The value m must be sufficiently small to apply the Cross-encoder and Reader models in the next step. The question is then paired with each retrieved passage to create the question-passage pairs. The pairs are fed to the Cross-Encoder model to obtain the relevance scores of m passages concerning the question. At the same time, the pairs are also fed to the Reader model to extract the answer spans for each passage. Only top n answer span predictions with the highest confidence score are kept. Finally, the obtained relevance score and answer span predictions are used to rerank the passages. Passages are sorted using the relevance score, and only the passages that contain at least one of the n answer spans are selected. When checking if a passage contains an answer span, both the passage and answer span are normalized by converting to lowercase and removing punctuation. This process results in n reranked passages. The detailed architectures of the Cross-Encoder and Reader models are described in Sections 4.5 and 4.6, respectively.

4.4. Retriever for multi-hop QA

We propose *ViWiQA-Multi Retriever*, a retriever method for Vietnamese multi-hop Open-domain QA. The retriever aims to retrieve the pair of passages that can be used to perform multi-hop reasoning and find the answer. Fig. 3 shows the process of the proposed method. Given a question, the same process of single-hop retriever described in Section 4.3 is carried out to retrieve top n passages $P_1, P_2, \dots, P_n$. The highest-ranking passage (P_1) is selected as the first passage in the passage pair. Let A_1 be the article containing the passage P_1 ; the c_1 articles connected to A_1 via hyperlinks $\{A_{1,1}, A_{1,2}, \dots, A_{1,c_1}\}$ are retrieved using the Wikipedia hyperlink graph. The passages of the connected articles are then collected and considered as candidates for the second passage in the pair. Using the candidate passages and the given question, the process of single-hop retriever is once again carried out to retrieve r passages $Q_1, Q_2, \dots, Q_r$. The highest-ranking passage (Q_1) is selected as the second passage for the passage pair. The final passage pair (P_1, Q_1) is then obtained.

In our experiments, only the highest-rank passage (P_1) is considered the first passage in the pair. As a result, only c_1 articles connected to P_1 in the hyperlink graph $\{A_{1,1}, A_{1,2}, \dots, A_{1,c_1}\}$ are used to get candidate passages for the second passage. However, it is also possible to consider lower-rank passages $P_2, \dots, P_n$ and their corresponding connected articles in the hyperlink graph as shown below.

$$P_2 \rightarrow \{A_{2,1}, A_{2,2}, \dots, A_{2,c_2}\}$$

$$P_3 \rightarrow \{A_{3,1}, A_{3,2}, \dots, A_{3,c_3}\}$$

...

$$P_n \rightarrow \{A_{n,1}, A_{n,2}, \dots, A_{n,c_n}\}$$

A strategy to choose the final passage pair among the candidate pairs will then be needed. We leave this to future work.

4.5. Cross-encoder model

To predict the relevance score for the question-passage pairs, the Cross-Encoder architecture (Devlin et al., 2019) with Transformer (Vaswani et al., 2017) is employed, following the work of Nogueira and Cho (2019). Fig. 4 shows the detailed architecture of the Cross-Encoder model. As the inputs are in Vietnamese, the pre-trained model XLM-RoBERTa (Conneau et al., 2020) is used as our Transformer model. The model architecture of XLM-RoBERTa is similar to the Transformer model. We highlight some key designs of the Transformer model and the pre-training technique of XLM-RoBERTa below. The complete architecture and pre-training process can be found in the work of Vaswani et al. (2017) and Conneau et al. (2020). The critical components of the Transformer model are the attention mechanism, specifically the Scaled Dot-Product Attention and the MultiHead Attention. We

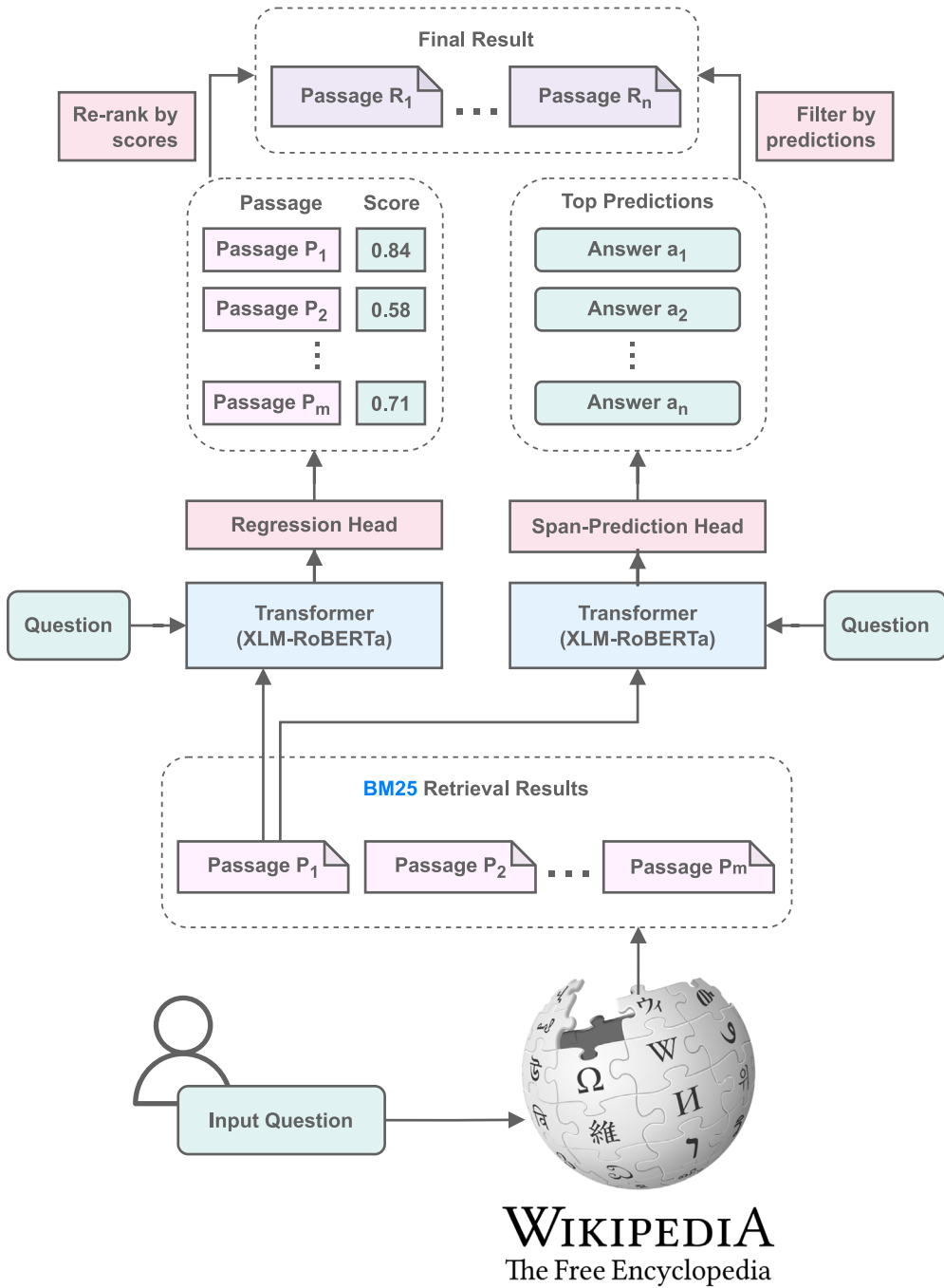


Fig. 2. Overall of the proposed single-hop retriever component of ViWiQA-Single.

outline the mathematical formulas from the Transformer paper below. Attention functions map a query vector (Q) and a collection of key-value vector pairs (K and V) to an output. The formula for Scaled Dot-Product Attention is as follows.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The formula for the MultiHead Attention is as follows.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

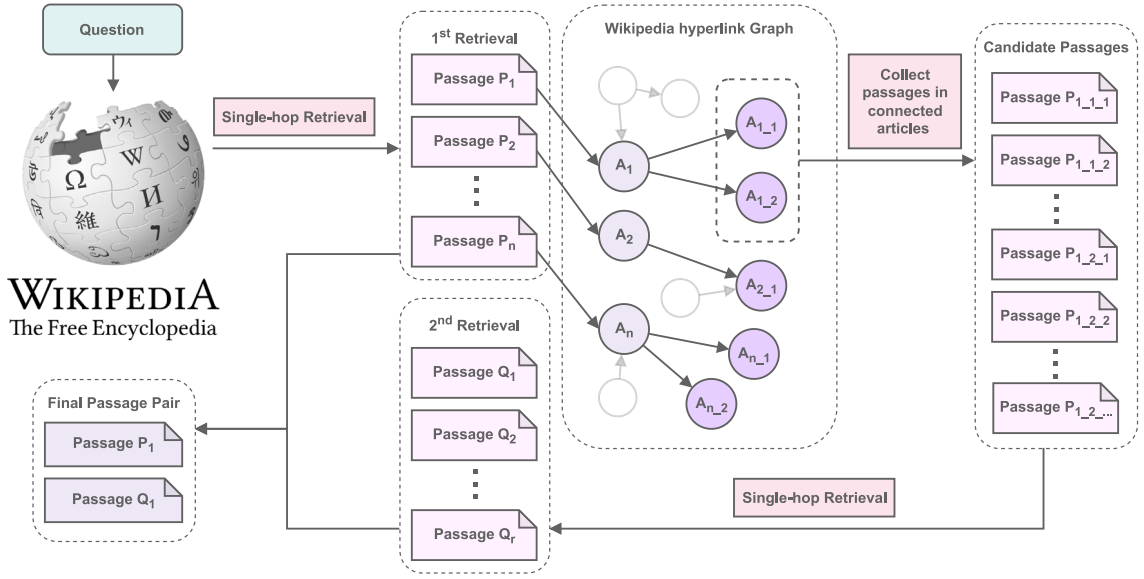


Fig. 3. Overall of the proposed multi-hop retriever component of ViWiQA-Multi.

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

W_i^Q, W_i^K, W_i^V are learnable parameter matrices

The attention mechanism is used in encoder–decoder layers for the decoder to attend over the output of the encoder. The mechanism is also used in the self-attention of the encoder and decoder layers. XLM-RoBERTa is pre-trained with a multilingual Mask Language Model objective on 100 languages, including Vietnamese. The tokenizer used XLM-RoBERTa employ Sentence Piece (Kudo & Richardson, 2018) with subword regularization (Kudo, 2018).

For predicting the relevance score between the question and the passage, the input question and passage are concatenated and separated by a *[SEP]* token. The classification token *[CLS]* is placed at the beginning, representing the sentence-level classification. Following the work of Devlin et al. (2019), we add a sequence regression head. The regression head outputs a score from 0 to 1, indicating the relevance of the passage with respect to the given question. We use the default configurations of the XLM-RoBERTa model where the last layer hidden-state of the classification token has the size of 768 and the drop-out rate of 0.1. The sequence regression head has the input size 768 and the output size 1, where the output indicates the regression output. An *Mean Square Error* (MSE) loss is calculated using the regression outputs and the labels. The label is 0.0 for negative examples and 1.0 for positive examples. The calculated MSE loss is used as the training objective to train the model. The MSE formula is given as follows.

$$\text{MSE}(\hat{y}, y) = \frac{1}{m} \sum_i (\hat{y} - y)^2$$

where m is the number of example inputs, $\hat{y}$ is the prediction of the model, and y is the regression target (label).

Alternatively, we can use the CrossEntropy loss instead of the MSE loss to train the model. When using the CrossEntropy loss, we define two classes indicating the relevance between the question and the passage: “Relevant” (denoted by label 1) and “Irrelevant” (denoted by label 0). The final layer of the model will output the two logits of these classes. The CrossEntropy loss is calculated using the following formula.

$$\text{CrossEntropyLoss}(\hat{y}, y) = - \frac{e^{\hat{y}_y}}{\sum_{c=1}^C e^{\hat{y}_c}}$$

where C is the number of classes, $\hat{y}$ is the prediction of the model, and y is the target class (0 or 1).

In the ablation study, we conduct experiments of Cross-Encoder with the MSE loss and CrossEntropy loss to compare the impact of these losses. The Cross-Encoder aims to output a score indicating the relevance between the question and the passage to rank the retrieval passages. The model trained with MSE loss will output a single score from 0 to 1, which is straightforward for ranking. On the other hand, CrossEntropy loss will output the two logits for the two classes “Relevant” and “Irrelevant”. We use the logit of the “Relevant” class as the score to rank the retrieval passages in the case of CrossEntropy loss. Because the model trained with MSE loss has a more straightforward way to output the relevance score, we use MSE loss in our main experiments.

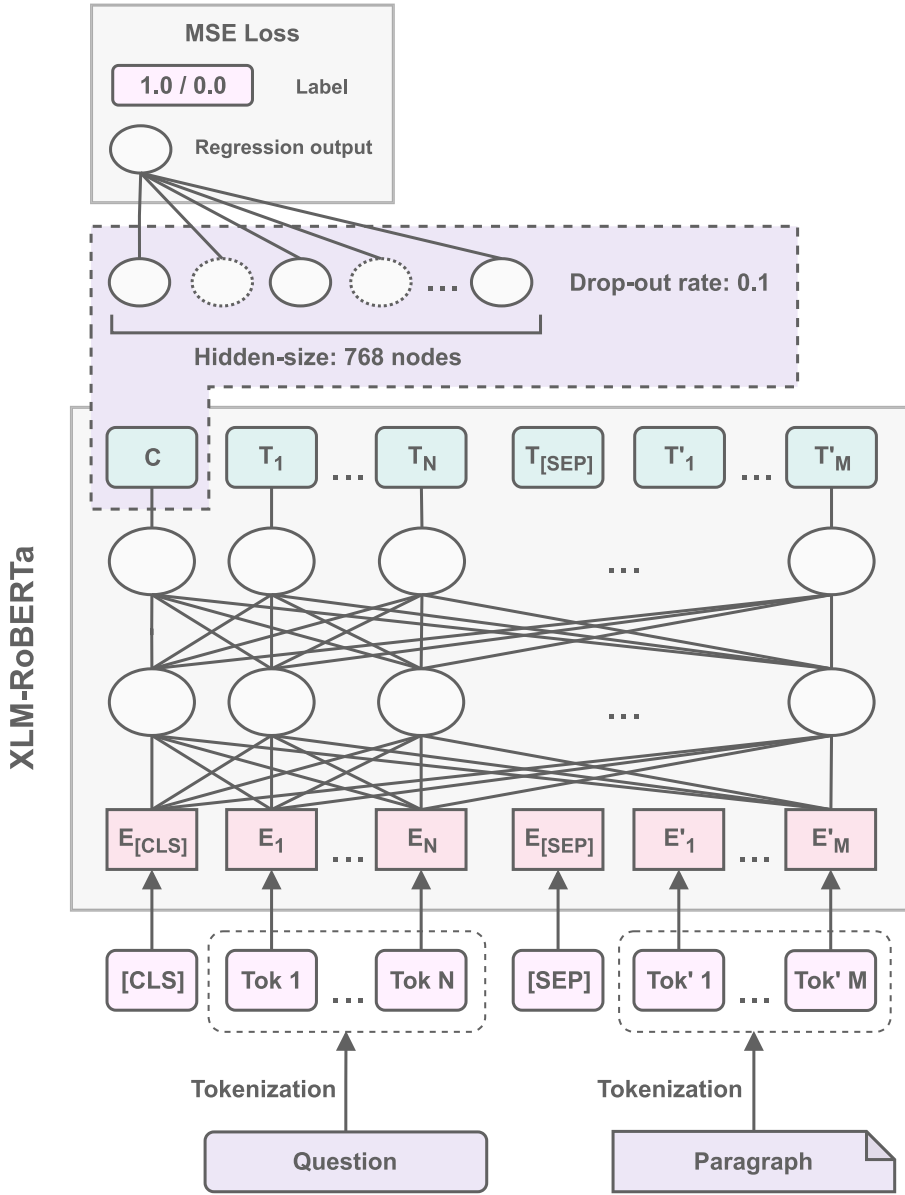


Fig. 4. Detailed architecture of Cross-Encoder model for predicting paragraph relevance scores.

4.6. Question-answering reader model

For the Reader model, we employ XLM-RoBERTa (Conneau et al., 2020) with a span classification head behind the final hidden-states to calculate the logits determining if a span is the start/end of an answer. Fig. 5 describes the detailed architecture of our Reader. We use the default configurations of the large version of the XLM-RoBERTa model. The two vectors $S \in \mathbb{R}^H$ and $E \in \mathbb{R}^H$ are introduced for calculating the start/end logits. The answer-start probability of token i is calculated using T'_i and S , formulated as follows.

$$Pstart_i = \frac{e^{S \cdot T'_i}}{\sum_j e^{S \cdot T'_j}}$$

A similar formula is applied for the end of the span.

$$Pend_i = \frac{e^{E \cdot T'_i}}{\sum_j e^{E \cdot T'_j}}$$

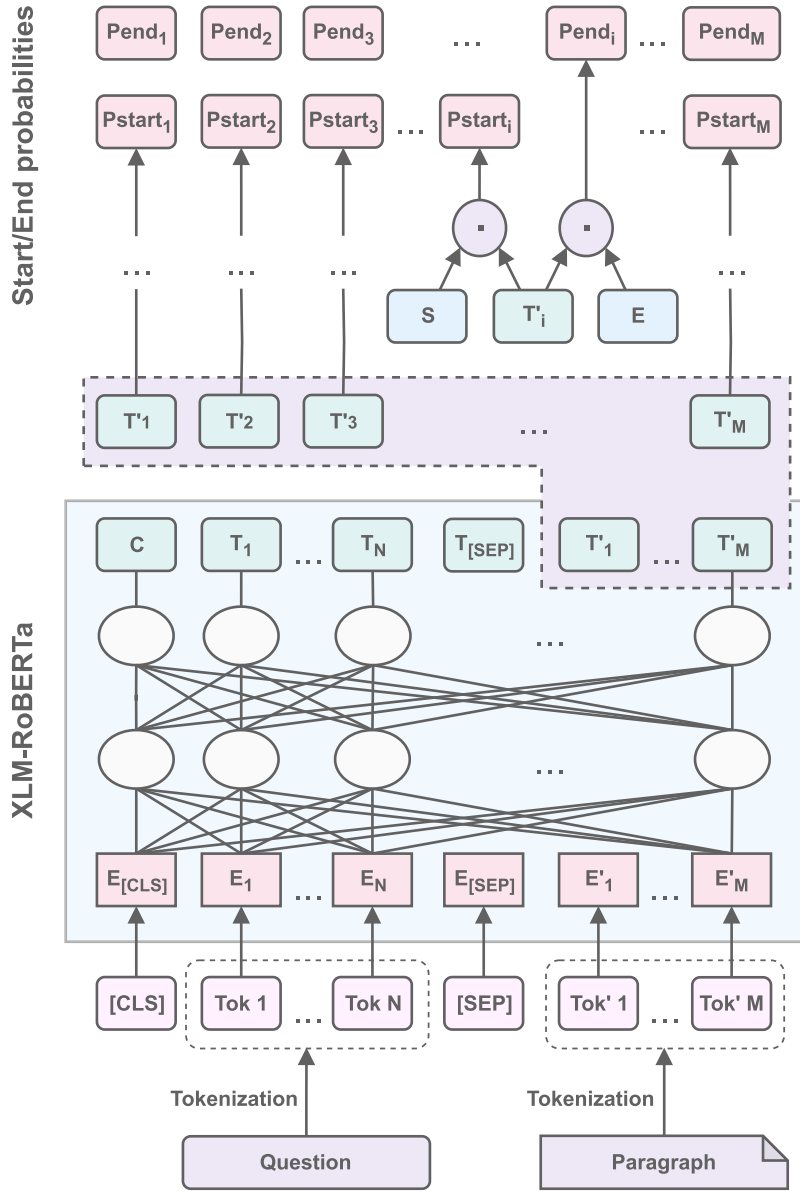


Fig. 5. Detailed architecture of Reader model for predicting answer spans.

A candidate answer spanning from i to j ($i \leq j$) has the score computed as $S.T'_i + E.T'_j$. The best-score span is selected as the predicted answer.

4.7. Training

The Cross-Encoder and the QA Reader in ViWiQA require training to adapt to the Vietnamese QA datasets. Negative and positive examples first need to be sampled to train the Cross-Encoder. Question-passage pairs where the passage holds the gold answer are considered positive examples, while the pairs with the passage irrelevant to the question are considered negative examples. We follow one negative sampling approach proposed by Karpukhin et al. (2020) that uses top passages retrieved by Lucene-BM25, which have many matched tokens in the question but do not hold the answer. For each question and the gold answer in the dataset, Lucene-BM25 is used to retrieve top n passages and traverse through n passages from the highest-ranked to the lowest-ranked passage. A passage containing the gold answer is marked as a positive example. Otherwise, it is marked as a negative example. We continue the process until the ratio between the negative and positive examples of the given question exceeds a predefined value r . In our experiments, we use $n = 100$ and $r = 7$. The choice of these parameters affects the training/inference times and the

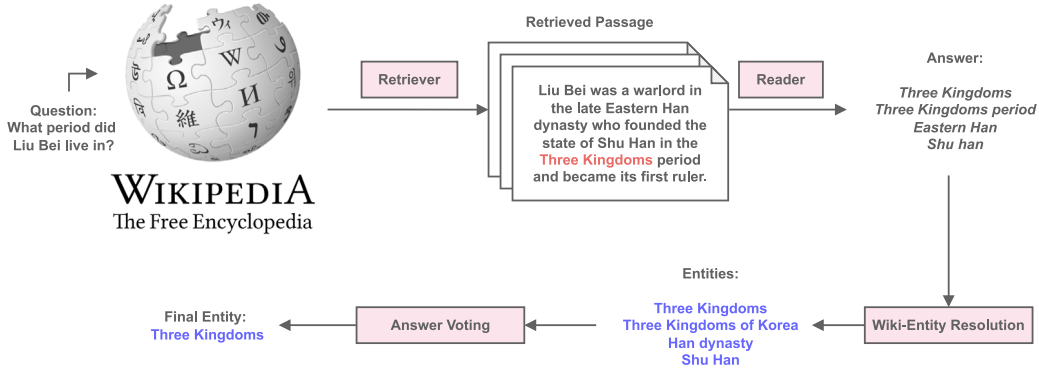


Fig. 6. Overall end-to-end QA system with Wikipedia entity resolution.

performance of the models. Nogueira and Cho (2019) use $n = 1000$ for MS MARCO, which is a much larger dataset than ViQuAD. In the DPR paper (Karpukhin et al., 2020), negative sampling with $r = 7$ shows good performance, while larger values of r do not show significant differences. Our preliminary study finds that $n = 100$ and $r = 7$ are suitable for balancing performance and efficiency. We train the Cross-Encoder on the sampled training data. The number of epochs trained is 3; The initial learning rate is $1e-5$; The batch size is 32.

The QA Reader of ViWiQA is trained using the annotated answer span start/end positions from the QA dataset. The number of epochs trained is 5; The initial learning rate is $1e-5$; The batch size is 16. We observed that the reader model converged at around epoch 2.5.

4.8. End-to-end QA system

There are two main modules in the end-to-end QA system: Retriever and Reader. Given a question, the Retriever retrieves the most relevant passage from the knowledge base, and the Reader reads the relevant passage to find the answer. We proposed two separate QA systems for single-hop and multi-hop QA, namely ViWiQA-Single and ViWiQA-Multi, respectively. The systems employed the corresponding retrievers shown in Sections 4.3 and 4.4.

We use the same Reader component, which has the architecture and training process shown in Sections 4.3 and 4.7, for both QA systems. For ViWiQA-Single, the question and the passage retrieved using the single-hop retriever are used as the input for the Reader. For ViWiQA-Multi, the question and the passage pair retrieved using the multi-hop retriever are used as the input for the Reader. Upon being fed to the Reader, the two passages in the pair are concatenated to form a single passage. Because the Vietnamese multi-hop QA dataset VIMQA (Le et al., 2022) has a similar format as HotpotQA (Yang et al., 2018) and contains Yes/No questions, a Yes/No tag in Vietnamese (“đúng/không”) is inserted at the beginning of the passage to enable the Reader to extract Yes/No answers.

4.9. Wikipedia-entity-resolution and model ensemble

In addition to ViWiQA, we also propose a simple approach to ER in QA and develop an entity-level end-to-end QA system (ViWiQA-ER) that accommodates the requirements of the Zalo AI Challenge 2022. In the Zalo AI Challenge contest, the End-to-end QA task over Wikipedia requires the output answer to be a Wikipedia entity, a specific date, or a number. Moreover, we propose an ensemble method for multiple retrievers and readers to boost the performance of the QA system. Fig. 6 shows the overall end-to-end QA system with Wikipedia entity resolution.

For questions where the answers are entities, the goal is to convert the plain text answer from the QA model to the corresponding Wikipedia entity. We employ the *redirect pages* metadata from Wikipedia for entity resolution. In Wikipedia, whenever a user accesses a *redirect page*, it will redirect the user to another Wikipedia page that refers to the same entity. For example, whenever the Wikipedia page “UK” is accessed, it will redirect the user to the “United Kingdom” page. This behavior of Wikipedia is possible thanks to the *redirect* metadata created by Wikipedia users when writing the articles. We extract the *redirect* metadata from the Wikipedia dump and build the *redirect dictionary* where each entry is in the form of $A \rightarrow B$ (A is redirected to B). The key A of each entry is converted to lowercase. We transform the plain text answer from the QA model to the corresponding Wikipedia entity by matching the answer with the key from *redirect dictionary* in the three following ways: (1) The answer is converted to lowercase (For example: “Isaac Newton” becomes “isaac newton”); (2) The answer is converted to lowercase and remove any punctuation (For example, “Isaac Newton”, becomes “isaac newton”); (3) Only the capital words in the answer are used and converted to lowercase (For example: “by Isaac Newton” becomes “isaac newton”). In addition, (1) is applied before (2) because there are Wikipedia entities that contain punctuation, such as internet top-level domain “.ca” and “.us”. If there is no matching key in the *redirect dictionary*, we use a simple BM25 approach to retrieve n closest Wikipedia entities from the list of all entities. In n entities, the entity with the lowest

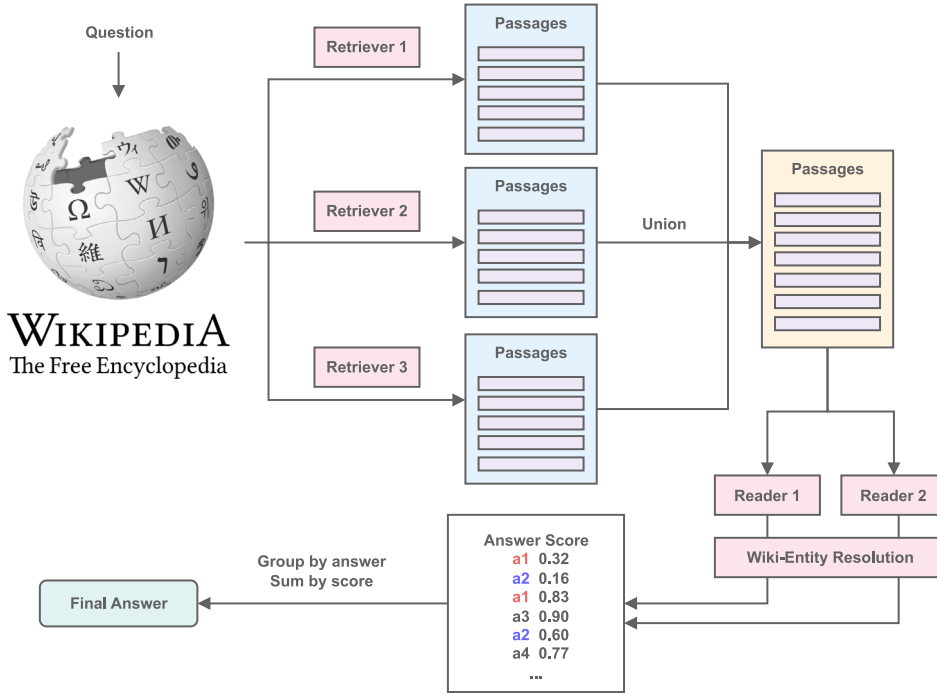


Fig. 7. Ensemble approach for multiple retrievers and readers.

Levenshtein distance to the answer is selected. For questions asking about specific dates or numbers, we employ *regular expression* to extract numbers and different date formats from the answers and rearrange them following the format from the QA task in the Zalo AI Challenge.

Model ensemble refers to techniques combining multiple models to produce better performance and plays an important role in an efficient end-to-end QA system. As our system consists of two separate steps, Retriever and Reader, we propose an ensemble method that combines multiple retrievers and readers to enhance the performance of the end-to-end system. Fig. 7 shows our ensemble approach. For a set of m retrievers $\{Retr_1, Retr_2, \dots, Retr_m\}$ and a set of n readers $\{Read_1, Read_2, \dots, Read_n\}$, the question is first passed to m retrievers to retrieve m set of passages. The union of these passages is then used to create the final retrieval result containing k passages. The retrieval result is then passed to n readers and the Wikipedia-entity resolution module to obtain $k * n$ answer entities along with their confidence score. The same entities are grouped together, and their scores are summed. Finally, the entity with the highest summed score is chosen to be the final answer entity.

5. Experimental result

In this section, we describe in detail the conducted experiments and analyze the results. Overall, we conduct experiments to compare the proposed methods with competitive baselines on different benchmarks, knowledge bases, and settings. The single-hop QA results show valuable insight into popular sparse/dense methods when adapted to Vietnamese and suggest that the proposed method performs better in retrieval and end-to-end metrics. The multi-hop QA results show that although dense methods have strong characteristics in locating correct articles, the proposed method has the best end-to-end performance, indicating some intriguing directions for future work combining the two approaches.

5.1. Experimental setup

5.1.1. Benchmarks and knowledge base

The proposed methods are evaluated on two large-scale human-generated Vietnamese QA Benchmarks, UIT-ViQuAD (Nguyen et al., 2020) for single-hop QA and VIMQA (Le et al., 2022) for multi-hop QA. The statistics of the two datasets are shown in Table 4.

Most of the experiments are conducted using the Vietnamese Wikipedia Corpus as a knowledge base for QA. After the pre-processing step, the Wikipedia Corpus contains approximately 1,200,000 articles and almost 4,000,000 passages. We also conduct experiments on UIT-ViQuAD knowledge base (Nguyen et al., 2022), a minimal subset of the Vietnamese Wikipedia knowledge base, to compare ViWiQA with the work of Nguyen et al. (2022). Table 5 compares the Vietnamese Wikipedia and UIT-ViQuAD knowledge base.

Table 4

Overall of Vietnamese datasets for single-hop QA (UIT-ViQuAD) and multi-hop QA (VIMQA).

Dataset	Train	Dev	Test	All
UIT-ViQuAD	18,579	2,285	2,210	23,074
VIMQA	8,041	1,003	1,003	10,047

Table 5

Comparison of Vietnamese Wikipedia and UIT-ViQuAD knowledge base.

	Vietnamese Wikipedia	UIT-ViQuAD
#articles	1,273,420	174
#passages	3,885,030	5,109

Table 6Top- k retrieval accuracy on UIT-ViQuAD development and test sets using Vietnamese Wikipedia knowledge base, measured by the proportion of top k passages retrieved containing the answer.

Method	Development set				Test set			
	Top-1	Top-5	Top-10	Top-20	Top-1	Top-5	Top-10	Top-20
Lucene-BM25 (Lin et al., 2021)	36.96	55.09	61.72	66.94	30.43	47.78	54.17	60.28
DPR (Karpukhin et al., 2020)	11.47	24.07	29.93	36.54	9.32	19.95	25.79	32.94
ColBERT (Santhanam, Khattab, Saad-Falcon, et al., 2022)	48.95	65.76	70.50	74.28	43.25	58.74	64.18	68.30
ViWiQA-Single Retriever (Ours)	56.80	71.64	74.58	76.29	50.41	63.95	67.44	69.75

The performance of end-to-end systems is measured by Exact Match (EM) and F1 score. We use the official evaluation script from SQuAD (Rajpurkar et al., 2016). The EM score is measured by the parts of the predictions that exactly match the labeled answer spans after normalization and is formulated as follows.

$$\text{EM} = \frac{\# \text{ exactly correct answers}}{\# \text{ questions}}$$

The F1 score is measured by Precision and Recall, which are formulated as follows.

$$\text{Precision} = \frac{\# \text{ correctly predicted tokens}}{\# \text{ predicted tokens}}$$

$$\text{Recall} = \frac{\# \text{ correctly predicted tokens}}{\# \text{ tokens in gold answer-span}}$$

$$\text{F1} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

The ViWiQA-ER is evaluated on the public test set of the Zalo AI Challenge 2022 (End-to-end Question-Answering task). The test set consists of approximately 600 questions whose answers are a Wikipedia entity, a specific date, or a number. The evaluation is measured by the EM score.

5.1.2. Competitive baselines

We adopt competitive methods in Open-domain QA to the Vietnamese QA datasets and use them as baselines for evaluation. For single-hop QA, the baselines include sparse retriever Lucene-BM25, dense retriever DPR (Karpukhin et al., 2020) and ColBERT (Khattab & Zaharia, 2020; Santhanam, Khattab, Saad-Falcon, et al., 2022). Lucene-BM25 has been proven to be a competitive baseline to evaluate Open-domain QA systems. Especially in datasets like SQuAD, UIT-ViQuAD, or VIMQA, the question and the passages have a high number of overlapping tokens, giving BM25 a benefit. DPR is a retriever method that uses dense representations and establishes competitive results on various English Open-domain QA Dataset such as NaturalQuestion (Kwiatkowski et al., 2019), Trivia (Joshi et al., 2017), WebQuestions (Berant, Chou, Frostig, & Liang, 2013), and TREC (Baudiš & Šedivý, 2015). ColBERT is a line of work in dense retrieval based on BERT and achieves competitive results in retrieval datasets such as MSMARCO (Nguyen et al., 2016) and TREC. Moreover, the results of DrQA (Chen et al., 2017), BERTSerini (Yang, Xie, et al., 2019), and XLMQA (Nguyen et al., 2022) on UIT-ViQuAD are taken from the work of Nguyen et al. (2022) to compare with ViWiQA on the UIT-ViQuAD knowledge base.

For multi-hop QA, the baselines include sparse retriever Lucene-BM25 and multi-hop dense text retrieval (MDR) (Xiong et al., 2021). MDR is a dense retrieval method designed for Open-domain multi-hop QA, which achieves competitive results on HotpotQA (Yang et al., 2018).

5.1.3. Implementation details

We employed Pyserini (Lin et al., 2021) to index the Vietnamese Wikipedia Corpus and build the Lucene-BM25 retriever. We follow the implementation of Wolf et al. (2020) to implement our Cross-Encoder and Reader models as described in Section 4.3. XLM-RoBERTa_{Base} and XLM-RoBERTa_{Large} are used for the Cross-Encoder and the Reader, respectively. For ViWiQA-Single, we let $m = 100$ be the number of passages retrieved by Lucene-BM25, and $n = 30$ be the number of top answer span predictions to keep.

Table 7

End-to-end single-hop QA performance on UIT-ViQuAD using the Wikipedia knowledge base. The Reader model of these systems is XLM-RoBERTa_{Large}.

System	Development set		Test set	
	EM	F1	EM	F1
Lucene-BM25	27.61	39.66	21.33	34.59
ColBERT	38.67	52.94	33.11	49.23
ViWiQA (Ours)	42.93	57.22	36.37	52.95

Table 8

End-to-end QA performance on UIT-ViQuAD test set using UIT-ViQuAD knowledge base.

System	EM	F1
DrQA (Chen et al., 2017)	17.87	37.37
BERTserini (Yang, Xie, et al., 2019)	36.52	55.55
XLMRQA (Nguyen et al., 2022)	47.96	61.83
ViWiQA-Single (Ours)	53.98	69.59

To adapt competitive baselines in English to Vietnamese, we employ the official implementation and code provided by the cited paper for DPR (Karpukhin et al., 2020), ColBERT (Santhanam, Khattab, Saad-Falcon, et al., 2022), MDR (Xiong et al., 2021). We follow the DPR paper to train its dual-encoders using the UIT-ViQuAD dataset and the Vietnamese Wikipedia knowledge base. Because DPR uses the in-batch negatives technique and the training batch sizes greatly impact the DPR performance, we experiment with different batch sizes and select the best results as the baseline. As the current DPR architecture only accepts a variant of BERT, we employ the pre-trained model Multilingual-BERT in our DPR adaptation to Vietnamese. For ColBERT and MDR, we use XLM-RoBERTa-Base as the backbone Transformer model.

5.2. Results

5.2.1. Single-hop question-answering

We conducted experiments to evaluate the proposed single-hop retriever and end-to-end QA system *ViWiQA-Single*. Table 6 shows a comparison of different retriever methods on UIT-ViQuAD development and test sets using Vietnamese Wikipedia knowledge base, measured by top- k accuracy ($k \in 1, 5, 10, 20$). The result indicates that *ViWiQA-Single* retriever outperforms significantly and consistently compared to Lucene-BM25 and DPR. The performance gap is especially large when k is small. The top-1 accuracy of our method is better than Lucene-BM25 by about 20%. The experiment also suggests that DPR suffers greatly in multilingual and low data situations, complementing the observation of Gao and Callan (2021) and Ren et al. (2021) about the limitation of dense representation via dual-encoders. The result also suggests that ColBERT performs noticeably better than BM25 and DPR when adapted to Vietnamese, although it is a dense retrieval approach. However, the performance of ColBERT is still lower than *ViWiQA-Single* in all top- k accuracy.

Table 7 shows the end-to-end evaluation on UIT-ViQuAD. The result suggests that better retrieval accuracy can enhance the end-to-end QA results and that *ViWiQA-Single* significantly outperforms Lucene-BM25 by about 15% and 18% absolute in EM and F1 scores, respectively.

We also conduct experiments over the UIT-ViQuAD knowledge base, a minimal subset of the Vietnamese Wikipedia knowledge base, to compare *ViWiQA-Single* with the work of Nguyen et al. (2022). Table 8 compares different systems over the UIT-ViQuAD knowledge base. The result of DrQA and BERTserini is derived from the research of Nguyen et al. (2022). The result indicates that *ViWiQA-Single* greatly outperforms other QA systems on the single-hop Vietnamese QA dataset.

5.2.2. Multi-hop question-answering

This section evaluates the proposed multi-hop retriever and end-to-end multi-hop QA system *ViWiQA-Multi*. Table 9 shows the multi-hop retrieval accuracy over the Vietnamese Wikipedia knowledge base. The accuracy is measured using three metrics: the percentage of retrieved passage pairs with at least one passage whose title is correct (1C), the percentage of retrieved passage pairs with two passages whose titles are correct (2C), and the percentage of retrieved passage pairs that contain the answer span excluding Yes/No questions (CA). A title is considered correct if it is the same as the title of the gold passage in the dataset. It is important to note that (1C) and (2C) only show that the title of the passage is correct and does not show that the content of the passage is the same as the gold passages. Because the retrieved passages are from the retrieval corpus, whose passages are 100-word chunks, it is not possible to compare the content of the retrieved passages with the gold passages. Therefore, we only compare the titles of the passages. Despite having lower accuracy in metric (1C) compared to Lucene-BM25, *ViWiQA-Multi* retriever has higher accuracy in metrics (2C) and (CA). The result indicates that the knowledge from the Wikipedia hyperlink graph successfully enhances the multi-hop retriever model in retrieving the correct passage pairs. In contrast, while Lucene-BM25 has a high percentage of retrieving one correct title in the passage pair, it struggles to find both correct titles. It is essential for multi-hop questions that both passages in the retrieved pair are correct so that the Reader model can perform multi-hop reasoning and find the answer. MDR results are

Table 9

Multi-hop retrieval accuracy over the Wikipedia knowledge base, evaluated on VIMQA dataset, measured in the percentage of retrieved passage pairs that have at least one correct title (1C), has two correct titles (2C) or contains the answer (CA). Yes/No questions are excluded when measuring CA accuracy. Note that (1C) and (2C) only show that the title of the passage is correct and does not show that the passage's content is correct.

Method	Development set			Test set		
	1C	2C	CA	1C	2C	CA
BM25	58.72	5.78	83.78	55.23	4.59	84.50
MDR	73.48	32.80	83.50	71.98	30.61	84.21
ViWiQA (Ours)						
Single	42.87	2.69	84.64	41.38	3.09	85.49
Multi	44.97	9.27	86.20	42.17	8.18	86.63

Table 10

End-to-end evaluation of multi-hop QA systems on VIMQA measured in Exact Match (EM) and F1 Score.

Method	Development set		Test set	
	EM	F1	EM	F1
Lucene-BM25	35.29	44.32	35.99	45.99
MDR	35.99	43.47	35.99	43.98
ViWiQA (Ours)				
ViWiQA-Single	38.78	50.10	39.08	51.42
ViWiQA-Multi	41.77	53.97	40.98	53.33

significantly higher than BM25 and ViWiQA in 1C and 2C, indicating its effectiveness in retrieving passages that belong to the correct article. However, it is important to note that (1C) and (2C) only show that the passages are in the correct article. These metrics do not show that the content of these passages is the same as the gold passages. Therefore, the end-to-end QA evaluation is essential to compare the effectiveness of these methods.

Table 10 shows the end-to-end evaluation on VIMQA. The result indicates that the proposed system significantly outperforms Lucene-BM25 with XLM-R Reader by about 5% to 6% in EM accuracy and 7% to 10% in F1 score. ViWiQA-Multi also performs better than ViWiQA-Single, indicating the improvement that the Wikipedia hyperlink graph has on the multi-hop retrieval process. Results in Table 9 and Table 10 show that although MDR retrieves more passages that belong to the correct articles, these passages do not facilitate the answer to the question. This suggests that MDR locates correct articles but retrieves the incorrect portion of the correct article. An approach to utilize the strong characteristics of MDR and ViWiQA-Multi is an intriguing and valuable direction for future work.

5.2.3. Wikipedia-entity-level question-answering

We conduct experiments on the ZALO AI Challenge 2022 (End-to-end Question-Answering task) public test set to evaluate the proposed ViWiQA-ER system. Table 11 shows the performance of ViWiQA-ER on the public test set, measured in EM score. The first group uses Lucene-BM25 (Lin et al., 2021) as the retriever, and the second group uses ViWiQA-Single Retriever as the retriever. Both groups use XLM-RoBERTa-Large (Conneau et al., 2020) as the reader. The result indicates the effectiveness that the proposed Wikipedia-Entity resolution module and the model ensemble method have on the performance of ViWiQA-ER. Approximately 8% and 10% absolute gain in EM score can be observed when applying Wikipedia-Entity resolution and Number/Date processing, respectively. The proposed ensemble method successfully boosts the system with about 2% absolute gain in EM score. Our ViWiQA-ER system achieved the **2nd Place** in the official **final result of ZALO AI Challenge 2022** (End-to-end Question-Answering task).

5.2.4. Ablation study

The first ablation study investigates the effectiveness of existing pre-trained models when applied to the Vietnamese QA reader. Table 12 compares the performance of different pre-trained Transformer reader models on the task of QA with given contexts. We compare pre-trained models on Vietnamese corpus PhoBERT (Nguyen & Tuan Nguyen, 2020) and multilingual corpus Multilingual-BERT (Devlin et al., 2019), XLM-RoBERTa (Conneau et al., 2020). The *Base* and *Large* versions of each pre-trained model are used, except for Multilingual-BERT, which is only available in *Base* version. The result suggests that XLM-RoBERTa is the most effective model when adapted to Vietnamese, outperforming the Vietnamese pre-trained model PhoBERT.

The second ablation study explores the in-batch negative technique of DPR when adapted to Vietnamese. Fig. 8 illustrates the influence of different training batch sizes on the Top-*k* retrieval accuracy of DPR. Owing to limited computational resources, the largest batch size in the experiment is 96. The result suggests that training batch size affects the performance of DPR to an extent. However, as shown in Section 5.2.1, the result at batch size 96 is still significantly lower than Lucene-BM25, indicating the DPR limitation when adapted to low-resource language.

Table 11
Evaluation of ViWiQA-ER on Zalo AI Challenge 2022 (End-to-end QA task) public test set.

Retriever	Reader	Wikipedia ER	Number/Date	Ensemble	EM
BM25	XLM-RoBERTa-Large				55.33
		✓			62.67
			✓		62.67
		✓	✓	✓	56.33
		✓	✓	✓	70.00
ViWiQA-Single	XLM-RoBERTa-Large				61.33
		✓			68.83
			✓		71.00
		✓	✓	✓	63.00
		✓	✓	✓	80.50

Table 12
Comparing different Vietnamese and multilingual pre-trained Transformer models on UIT-ViQuAD test set (QA with given contexts task).

Reader	EM	F1
PhoBERT _{Large}	64.17	82.45
PhoBERT _{Base}	60.19	79.56
Multilingual-BERT _{Base}	61.19	81.02
XLM-RoBERTa _{Large}	72.83	89.73
XLM-RoBERTa _{Base}	66.94	85.19

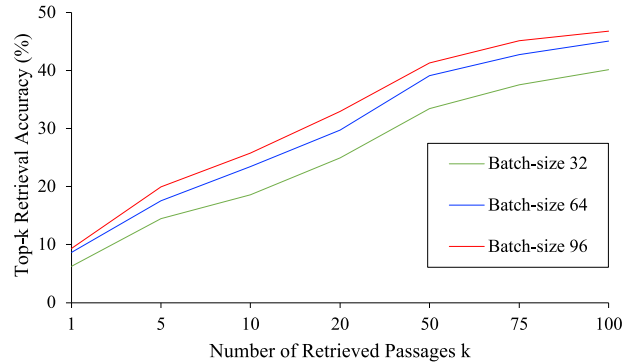


Fig. 8. Retrieval accuracy of DPR on UIT-ViQuAD test set at different training batch sizes when adapted to Vietnamese Wikipedia, measured by the proportion of top k retrieval results containing the answer.

We perform an ablation study focusing on analyzing the method in the ViWiQA-Single retriever. Table 13 shows the performance of ViWiQA-Single retriever in three different settings. The first setting only uses the relevance scores from the Cross-Encoder model to rerank the passages. The second setting only uses the top answer span predictions from the Reader model to filter the passages. The third setting is the full method where both Cross-Encoder and Reader outputs are considered, as in Section 4.3. Table 15 shows the evaluation of the end-to-end ViWiQA-Single in the same three settings. The result suggests that reranking passages using the Cross-Encoder model gives more performance gains than filtering passages using the Reader model. However, filtering passages using the Reader model still gives an essential improvement in the full method.

Because there are two choices of the loss function in the Cross-Encoder, we conduct experiments to compare the performance of these choices. Table 14 shows the top- k retrieval accuracy of the Cross-Encoder model trained using MSE loss and CrossEntropy loss. The results suggest no significant difference between the performance of these losses, although CrossEntropy loss shows a slight increment in the accuracy. For consistency, we use MSE loss for all other experiments.

We also analyze specific retrieval results from Cross-Encoder and ViWiQA-Multi Retriever to grasp the merits and demerits of the models. Tables 16 and 17 show two retrieval examples using Cross-Encoder and ViWiQA-Multi. In these examples, the models aim to retrieve the passage pair to support answering the multi-hop question. The retrieval results indicate that although Cross-Encoder can retrieve relevant passages to the question, these passages are not connected in a way that facilitates multi-hop reasoning. As a result, the passage-pair from Cross-Encoder cannot support answering the multi-hop question. On the other hand, between the first and the second passage in the passage-pair retrieved by ViWiQA-Multi Retriever, there is a connection through *bridge* entity like “Manchester United F.C” and “Warner Bros”. The *bridge* entities in multi-hop QA are usually entities that connect the contexts and

Table 13

Top- k retrieval accuracy on UIT-ViQuAD development and test sets using Vietnamese Wikipedia knowledge base, measured by the proportion of top k retrieval results containing the answer. ViWiQA-Single is evaluated in three settings: Only rerank using the Cross-Encoder (CE), only filter by the predictions of Reader (RE), and full method using both the CE scores and Reader predictions (Full). See text for more details.

Method	Development set				Test set			
	Top-1	Top-5	Top-10	Top-20	Top-1	Top-5	Top-10	Top-20
ViWiQA-Single Retriever (CE)	55.62	70.72	74.41	76.25	49.18	63.09	66.76	69.25
ViWiQA-Single Retriever (RE)	39.51	58.65	64.49	70.54	32.70	51.18	57.79	64.18
ViWiQA-Single Retriever (Full)	56.80	71.64	74.58	76.29	50.41	63.95	67.44	69.75

Table 14

Top- k retrieval accuracy of Cross-Encoder model when using Mean Squared Error (MSE) loss and CrossEntropy loss, measured by the proportion of top- k passages retrieved containing the answer.

Method	Development set				Test set			
	Top-1	Top-5	Top-10	Top-20	Top-1	Top-5	Top-10	Top-20
MSE Loss	55.62	70.72	74.41	76.25	49.18	63.09	66.76	69.25
CrossEntropy Loss	58.25	71.60	74.58	76.21	49.82	63.99	67.53	69.75

Table 15

End-to-end single-hop QA performance on UIT-ViQuAD using the Wikipedia knowledge base. ViWiQA is evaluated using only the Cross-Encoder (CE), only Reader predictions (RE), and full method (Full).

System	Development set		Test set	
	EM	F1	EM	F1
ViWiQA (CE)	42.14	55.98	35.46	51.64
ViWiQA (RE)	29.28	42.15	22.74	37.29
ViWiQA (Full)	42.93	57.22	36.37	52.95

facilitate the creation of multi-hop question (Le et al., 2022; Yang et al., 2018). Therefore, ViWiQA-Multi Retriever produces high-quality passage-pairs that can answer multi-hop questions. These examples indicate the effectiveness of using Wikipedia hyperlink graphs to select candidate passages that are connected to the first passage for the second passage retrieval.

6. Discussion

6.1. Theoretical implications

This study introduces new state-of-the-art retriever methods for Vietnamese Open-domain QA, which can be trained efficiently with low resources. Using these retrievers, we developed end-to-end Open-domain QA systems with state-of-the-art performance on Vietnamese Open-domain QA benchmarks. This study has the following theoretical implications. First, the experiments show that while dense retrievers architecture suffers greatly in multilingual and low data situations, cross-encoder can be trained efficiently with less amount of data. This finding validates previous observations of Gao and Callan (2021) and Ren et al. (2021). Secondly, we show that the hyperlink graph from Wikipedia is a valuable resource for enhancing multi-hop retrievers. This finding is in line with the work of Asai et al. (2020) and Zhu, Pang, et al. (2021). Different from previous works, for the single-hop retriever, we combine the two cross-encoder architectures, one with a regression head for reranking results and one with a span-classification head for filtering results, making the method robust in low data situations and achieve high performance; For multi-hop retriever, we use the hyperlink graph to obtain the candidates for the single-hop retriever, allowing the retrieval of correct passage pairs.

6.2. Practical implications

To the best of our knowledge, this work is the first to address the single-hop, multi-hop, and ER problems in Vietnamese Open-domain QA. The proposed system outperforms previous Vietnamese QA systems and achieved the Runner-up Prize in the Zalo AI Challenge 2022 Contest, indicating the potential for practical applications. This study has important practical implications and supports the research community in studying Vietnamese question-answering further.

6.3. Future work

In future work, we aim to enhance the capabilities of our Vietnamese open-domain question-answering (QA) system by addressing the current limitation and exploring the potential of large language models.

The current ViWiQA-Multi only output one passage pairs for multi-hop reasoning. We intend to explore different strategies to choose the final passage pair among the candidate pairs and conduct more ablation studies on our multi-hop retriever. Moreover,

Table 16

Example 1: Passage-pair retrieval using Cross-Encoder and ViWiQA-Multi. English translation is provided in *italic*.

Question: Biệt danh của một trong những câu lạc bộ ở Anh mà Fabien Alain Barthez từng chơi và đoạt huy chương là gì? (*What is the nickname of one of the clubs in England where Fabien Alain Barthez played and won a medal?*)

Answer: Quỷ đỏ (*The Red Devils*)

Cross-encoder retrieval

Title 1: Fabien Barthez

Passage 1 (From Wikipedia): Fabien Alain Barthez (; sinh ngày 28 tháng 6 năm 1971) là một cựu cầu thủ bóng đá người Pháp. Ông đã từng đoạt một số huy chương khi chơi ở vị trí thủ môn cho Marseille, Manchester United và cùng với đội tuyển bóng đá quốc gia Pháp giành chức vô địch tại World Cup 1998, Euro 2000 và lọt vào trận chung kết World Cup 2006. Ông cùng với Peter Shilton là 2 thủ môn giữ kỷ lục giữ sạch lưới nhất trong giải vô địch bóng đá thế giới, trong 10 trận. Ở câu lạc bộ, ông (*Fabien Alain Barthez (; born 28 June 1971) is a French former footballer. He won several medals while playing as a goalkeeper for Marseille, Manchester United and won the 1998 World Cup, Euro 2000, and reached the 2006 World Cup final with the France national football team. He and Peter Shilton are the two goalkeepers who hold the record for keeping the most clean sheets in the World Cup, in 10 matches. At club level, he*)

Title 2: West Ham United F.C.

Passage 2 (From Wikipedia): West Ham United Football Club là một câu lạc bộ bóng đá chuyên nghiệp Anh đặt trụ sở vùng phía đông thành phố London, thủ đô nước Anh. West Ham United đã 3 lần đoạt Cúp FA, 1 lần đoạt Cúp C2 châu Âu và 1 lần đoạt cúp Intertoto. Sân nhà của câu lạc bộ là sân vận động Olympic với sức chứa khoảng 60.000 khán giả. Biệt danh của câu lạc bộ là "The Irons" hoặc "The Hammers". Các đối thủ truyền thống của West Ham United là các câu lạc bộ cùng thành phố (*West Ham United Football Club is an English professional football club based in the east of London, the capital of England. West Ham United has won the FA Cup three times, the European Cup once, and the Intertoto Cup once. The club's home ground is the Olympic Stadium with a capacity of about 60,000 spectators. The club's nickname is "The Irons" or "The Hammers". West Ham United's traditional rivals are clubs from the same city*)

ViWiQA-Multi retrieval (ours)

Title 1: Fabien Barthez

Passage 1 (From Wikipedia): Fabien Alain Barthez (; sinh ngày 28 tháng 6 năm 1971) là một cựu cầu thủ bóng đá người Pháp. Ông đã từng đoạt một số huy chương khi chơi ở vị trí thủ môn cho Marseille, Manchester United và cùng với đội tuyển bóng đá quốc gia Pháp giành chức vô địch tại World Cup 1998, Euro 2000 và lọt vào trận chung kết World Cup 2006. Ông cùng với Peter Shilton là 2 thủ môn giữ kỷ lục giữ sạch lưới nhất trong giải vô địch bóng đá thế giới, trong 10 trận. Ở câu lạc bộ, ông (*Fabien Alain Barthez (; born 28 June 1971) is a French former footballer. He won several medals while playing as a goalkeeper for Marseille, Manchester United and won the 1998 World Cup, Euro 2000, and reached the 2006 World Cup final with the France national football team. He and Peter Shilton are the two goalkeepers who hold the record for keeping the most clean sheets in the World Cup, in 10 matches. At club level, he*)

Title 2: Manchester United F.C.

Passage 2 (From Wikipedia): Anh, giải đấu hàng đầu trong hệ thống bóng đá Anh. Với biệt danh "Quỷ Đỏ", câu lạc bộ được thành lập dưới tên Newton Heath LYR Football Club vào năm 1878, đổi tên thành Manchester United vào năm 1902 và chuyển đến sân vận động hiện tại, Old Trafford, vào năm 1910. Manchester United là một trong những câu lạc bộ thành công nhất tại Anh, giữ kỷ lục 20 lần vô địch bóng đá Anh, đoạt 12 Cúp FA, 5 Cúp Liên đoàn và giữ kỷ lục 21 lần đoạt Siêu cúp Anh. Câu lạc bộ đã giành (*England, the top league in the English football system. Nicknamed "the Red Devils", the club was founded as Newton Heath LYR Football Club in 1878, renamed Manchester United in 1902 and moved to its current stadium, Old Trafford, in 1910. Manchester United is one of the most successful clubs in England, holding a record 20 times English football championship, won 12 FA Cups, 5 League Cups, and holds the record of 21 times won the English Super Cup. The club won*)

the performance of the current single-hop retriever depends largely on the Cross-Encoder relevant scores. Although the phase of filtering retrieval results by the predictions of Reader shows notable improvement, more study is needed to increase the impact of this phase in the final result.

Large language models have shown significant progress recently with the proposal of many models in English, such as Bloom (Workshop et al., 2023), LLaMa (Touvron et al., 2023), T0 (Sanh et al., 2022), and multilingual models, such as Bloomz and mT0 (Muennighoff et al., 2022). These models show great potential and achieve competitive results in zero-shot settings on many benchmarks. Some work also proposes a new paradigm for Open-domain QA using generators rather than retrievers. However, work has yet to be conducted in exploring and adapting these approaches to Vietnamese Open-domain QA. In future work, we intend to investigate the integration of these large-scale pre-trained models and other paradigms rather than Retriever-Reader.

7. Conclusions

This study proposes effective single and multi-hop retriever methods for Open-domain QA in Vietnamese over the Wikipedia knowledge base. The single-hop retriever utilizes relevance scores produced by the Cross-Encoder model and answer predictions from the Reader model to enhance the retrieval. The multi-hop retriever enhances the quality of retrieved passage pairs by integrating the

Table 17

Example 2: Passage-pair retrieval using Cross-Encoder and ViWiQA-Multi. English translation is provided in *italic*.

Question: Công ty phát hành album Death Magnetic có trụ sở ở đâu? (*Where is the company that publishes the album Death Magnetic based?*)

Answer: Burbank

Cross-encoder retrieval

Title 1: Death Magnetic

Passage 1 (From Wikipedia): Death Magnetic là album phòng thu thứ 9 của ban nhạc heavy metal đến từ Mỹ Metallica, phát hành ngày 12 tháng 9 năm 2008 bởi Warner Bros. Records. Đây là album đầu tiên của nhóm có sự góp mặt của tay Bass Robert Trujillo, và nhà sản xuất Rick Rubin. Đây cũng là album phòng thu đầu tiên của Metallica hợp tác với Warner Bros. Records. Album này sau khi phát hành đã leo lên vị trí số 1 tại bảng xếp hạng Billboard 200 của Mỹ với 490.000 bản được tiêu thụ ngay tuần đầu tiên. Với thành (*Death Magnetic is the ninth studio album by American heavy metal band Metallica, released on September 12, 2008 by Warner Bros. Records. This is the group's first album to feature bassist Robert Trujillo, and producer Rick Rubin. This is also Metallica's first studio album in collaboration with Warner Bros. Records. This album after its release climbed to No. 1 on the US Billboard 200 chart with 490,000 copies sold in the first week. With this achievement.*)

Title 2: Danh sách album quán quân năm 2008 (Mỹ) (*List of number-one albums of 2008 (USA)*)

Passage 2 (From Wikipedia): 2009. Cô là nữ ca sĩ nhạc đồng quê duy nhất đạt được thành tích này trong lịch sử bảng xếp hạng Billboard 200. Hiện Swift được xếp hạng 5 trong danh sách các nữ nghệ sĩ hát đơn có được album quán quân lâu nhất, ngang hàng với Mariah Carey và Whitney Houston. Một vài album quán quân khác trong vài tuần bao gồm "Sleep Through the Static" bởi Jack Johnson và album thứ 9 "Death Magnetic" của Metallica; cả hai đều đứng đầu bảng trong 3 tuần liên tiếp. Trong năm 2008 có ba album nhạc phim (soundtrack) (*2009. She is the only female country singer to achieve this feat in the history of the Billboard 200 chart. Currently, Swift is ranked 5th on the list of female solo artists with the longest number-one album, equal to Mariah Carey and Whitney Houston. A few other number-one albums within a few weeks include "Sleep Through the Static" by Jack Johnson and the ninth album "Death Magnetic" by Metallica; both topped the table for 3 consecutive weeks. In 2008 there were three soundtrack albums (soundtrack)*)

ViWiQA-Multi retrieval (ours)

Title 1: Death Magnetic

Passage 1 (From Wikipedia): Death Magnetic là album phòng thu thứ 9 của ban nhạc heavy metal đến từ Mỹ Metallica, phát hành ngày 12 tháng 9 năm 2008 bởi Warner Bros. Records. Đây là album đầu tiên của nhóm có sự góp mặt của tay Bass Robert Trujillo, và nhà sản xuất Rick Rubin. Đây cũng là album phòng thu đầu tiên của Metallica hợp tác với Warner Bros. Records. Album này sau khi phát hành đã leo lên vị trí số 1 tại bảng xếp hạng Billboard 200 của Mỹ với 490.000 bản được tiêu thụ ngay tuần đầu tiên. Với thành (*Death Magnetic is the ninth studio album by American heavy metal band Metallica, released on September 12, 2008 by Warner Bros. Records. This is the group's first album to feature bassist Robert Trujillo, and producer Rick Rubin. This is also Metallica's first studio album in collaboration with Warner Bros. Records. This album after its release climbed to No. 1 on the US Billboard 200 chart with 490,000 copies sold in the first week. With this achievement.*)

Title 2: Warner Bros.

Passage 2 (From Wikipedia): Công ty Giải Trí Warner Brothers (hay Warner Bros., Warner Bros. Pictures) là một trong những hãng sản xuất phim và truyền hình lớn nhất thế giới. Đây là một chi nhánh từ Time Warner, trụ sở đặt tại Burbank, California và New York City. Warner Bros. có vài công ty con khác như Warner Bros. Studios, Warner Bros. Pictures, Warner Bros. Games, Warner Bros. Television, Warner Bros. Animation, Warner Home Video, DC Comics và New Line Cinema. Warner chiếm lĩnh một nửa thị trường The CW Television Network. Được thành lập năm 1918 bởi những người nhập cư từ (*Warner Brothers Entertainment Company (or Warner Bros., Warner Bros. Pictures) is one of the largest film and television production companies in the world. This is an affiliate from Time Warner, with headquarters in Burbank, California and New York City. Warner Bros. has several other subsidiaries such as Warner Bros. Studios, Warner Bros. Pictures, Warner Bros. Games, Warner Bros. Television, Warner Bros. Animation, Warner Home Video, DC Comics, and New Line Cinema. Warner dominates half of The CW Television Network's market. Founded in 1918 by immigrants from*)

Wikipedia hyperlink graph in the retrieval process. Using the proposed retrievers, we develop end-to-end Open-domain QA systems that achieve new state-of-the-art results in standard Vietnamese single and multi-hop QA datasets. The efficacy of the proposed systems can be confirmed by comparing the experimental results with strong baselines in resource-rich languages.

CRedit authorship contribution statement

Dieu-Hien Nguyen: Conceptualization, Methodology, Software, Writing – original draft. **Nguyen-Khang Le:** Conceptualization, Methodology, Software, Visualization, Writing – review & editing. **Le-Minh Nguyen:** Supervision, conceptualization, Writing – review & editing, Funding acquisition.

Data availability

Data will be made available on request.

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